Predicting Food Delivery Time

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Introduction

This data set prompted the scientific question: How does a range of predictors affect the final food delivery time? Found at *Kaggle.com,* the simulated dataset, https://www.kaggle.com/datasets/denkuznetz/food-delivery-time-prediction, reflects the real-world circumstances of food delivery.

Data Description

The dataset, which is simulated, was posted in January 2025. The dataset contains continuous predictors (distance, in kilometers, courier experience in years, and preparation time in minutes), discrete (order ID), as well as categorical (weather, traffic level, time of day, and vehicle type). A model fitting process will be conducted using these variables to predict the continuous target variable: delivery time, in minutes. After learning from the dataset, with our main goal for the analysis in mind, more specific questions arose about individual predictors and their influence on company efficiency. The amount of influence the predictor, vehicle type, will have on delivery time became apparent since bikes, scooters, and cars travel at vastly different rates. Can the approximate distances be covered in a similar amount of time by all vehicle types given the variability in other factors such as weather and traffic level? Another aspect that was questioned was the food delivery company's relationship with its partners. Does time of day have a strong influence on preparation time since restaurants tend to be busier during the lunch and dinner rush? One last question that emerged stemmed from the need to ensure the efficiency of the drivers: Does delivery time have a sensible correlation with distance? Results from the analysis will be critical in optimizing food delivery industry operations to identify key causes of delay in delivery, which can be used to optimize customer experience.

Data Cleaning

Model Building Process

Of the Six Steps of Model-Building, the group has defined the scientific problem, as well as, acquired and cleaned the dataset in preparation for exploratory data analysis (EDA). The cleaning process involved removing the Order_ID variable, which is only used for organizing data. Additionally, we identified missing numeric

Implemented Algorithms

One algorithm we have implemented is a basic linear regression model. From the basic diagnostic plots, linear regression may not be the best possible model to represent the data. The Q-Q Residuals plot particularly shows a large deviation on the right-most tail. Additionally, the distribution is slightly positively skewed and violates normality.